

Dipanshu Choudhary

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SUMMARY

Software Engineer specializing in **AI/ML and backend systems**, with hands-on experience building production applications spanning LLMs, RAG, agentic workflows, real-time APIs, and asynchronous processing. Strong foundation in **Python, Java, FastAPI, React/Next.js, PostgreSQL, Redis, and system design**, with practical experience in security, evaluation, reliability, and scalable application architecture.

EXPERIENCE

Junior Full Stack AI Engineer Intern

Onsite

Namekart (May 2026 - July 2026)

- Engineered a production **domain-evaluation pipeline** processing **~150,000 domains/day** and ranking a high-confidence buy list of **~2,000 Strong domains** (capped at **~3,000**) through multi-stage quality filters, ranking algorithms, and domain-quality parameters.
- Built **evaluation and benchmarking pipelines** to measure automated decision accuracy against shortlist outcomes, iterating scoring, prompts, and decision rules to improve precision of the **~2,000-domain Strong set**.
- Optimized the internal operations dashboard and automated **CBFR (Company-Based Funding Research)** with parallel workflows that research funded startups, classify domains as **L0/L1**, suggest brand domains, and find founder/CEO contacts, enabling processors to review and submit in Workstation instead of researching from scratch.

Freelancer - AI Product Engineer

Remote

TZURONI LTD. (Mar 2026 - Apr 2026)

- Engineered a production market-intelligence platform with data ingestion, preprocessing, semantic search, and vector-based retrieval across SEC filings and social media data, adding sentiment analysis and context-aware LLM insights.
- Built modular backend workflows for scalable ingestion and conversational analytics, with resilient fallback mechanisms, workflow orchestration, and RAG-powered retrieval.

PROJECTS

Quizzer - Secure Assessment Platform

Next.js, TypeScript, FastAPI, PostgreSQL, Redis, Celery, LangGraph, LangSmith, Ragas, DeepEval, Docker | Dec 2025 - Present | [Live](#)

- Architected a **dual-runtime assessment platform** with cookie-JWT creator workspace and isolated X-Attempt-Token exam runtime, spanning **~200 REST endpoints across 31 routers, 2 WebSockets, and 35 data models** for authoring, secure delivery, live monitoring, Arena battles, grading, analytics, Google Classroom/Calendar/Drive, and Microsoft Teams.
- Built a **Celery-based content-to-exam pipeline** ingesting PDF/DOCX/PPTX/images via Tesseract OCR, text, URLs, YouTube captions, and Drive files; supports **5 question types** with a multi-provider AI Gateway, **12 routed models, 3 plan tiers, and Redis-backed multi-key failover**.
- Engineered **server-authoritative exam security** with Redis TTL timers (**300s floor**), single-session locks, **60/min autosave, 13 violation types**, browser guards, TensorFlow.js/COCO-SSD phone/face signals, WebSocket integrity events, and **35 Redis rate-limit policies across 4 algorithms**.
- Built **QUE as a separate intent-aware AI service** with **20 tools, 10 zero-LLM workflow cards, hybrid Chroma + BM25 RRF retrieval, fail-closed role checks, and propose-then-confirm write actions**, backed by **170+ offline evaluation cases** covering retrieval, tools, permissions, follow-ups, paraphrases, and prompt injection.

Life Route AI - Emergency Healthcare Coordination

FastAPI, PostgreSQL, pgvector, Redis Streams, WebSockets, Event-Driven Architecture, Geospatial Routing, Docker | June 2026 - Present | [GitHub](#)

- Architected a **real-time healthcare emergency coordination platform** spanning **20+ REST/WebSocket APIs and 15+ data models** across patient identity, Health Passport, emergency events, ambulance telemetry, hospital capability/capacity, care paths, referrals, transfers, and follow-up, connecting five stakeholders across a **patient → bystander → ambulance → hospital → clinician** workflow; designed for **99.9%+ availability, 100+ concurrent telemetry streams, and 1,000+ events/sec**.
- Engineered an **event-driven AI triage and care-path orchestration system** using LangGraph, FastAPI, PostgreSQL/pgvector, Redis, and LLMs, combining deterministic safety rules with selective AI reasoning; processed **10+ real-time patient signals (HR, SpO₂, BP, trends, symptoms, location, medical context)**, with targets of **<500 ms p95 geospatial hospital retrieval and <1 sec p95 care-path ranking** excluding external network latency.
- Built a real-time emergency coordination layer connecting voice and wearable alerts with ambulance telemetry, hospital notifications, rerouting, and digital handoffs; targeted **<200 ms p95 critical API latency, <100 ms p95 telemetry propagation, <1 sec vital visibility, <2 sec detection-to-workflow initiation, <5 sec hospital pre-alert delivery, and <3 sec route reassessment**, with **<10 ms Redis event latency and end-to-end auditability**.

SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript, Go

Software Engineering: Data Structures & Algorithms, OOP, System Design, REST APIs, WebSockets, Caching, Rate Limiting, Asynchronous Processing

Backend & Full Stack: Next.js, React, FastAPI, PostgreSQL, MySQL, MongoDB, Redis, Celery

Cloud & DevOps: Docker, Kubernetes, GCP, Git, GitHub

AI Engineering: Generative AI, LLMs, RAG, Agentic AI, LangChain, LangGraph, CrewAI, Prompt Engineering

LLM Evaluation: Custom Offline Evals, Observability, LangSmith, Ragas, DeepEval

Machine Learning: Deep Learning, Transformers, NLP, TensorFlow, PyTorch

ACHIEVEMENTS

- AI INNOVATION 2026 Winner** — Microsoft × Kyndryl
- Clean Coder 2026** — Google

EDUCATION

Amity University, Noida

Bachelor of Computer Applications (BCA)

2024 - 2027 (Expected)

CGPA: 8.7/10